

Estimation in Statistics

Estimation is a statistical process used to infer the value of an unknown population parameter based on sample data. It helps us make informed decisions without having to survey the entire population. Estimation is broadly classified into two types:

1. **Point Estimation**
2. **Interval Estimation**

1. Point Estimation

A point estimate is a single numerical value used to approximate a population parameter. It is derived from sample data using statistical formulas.

Example:

Suppose we want to estimate the **average height of all students in a school**. Measuring the height of every student is impractical, so we take a **random sample** of 50 students and calculate their **sample mean height** (e.g., 160 cm).

Here, the sample mean ($\bar{x} = 160 \text{ cm}$) is the point estimate of the population mean (μ).

Commonly used point estimators:

- **Sample Mean ($\bar{x}$)** → Estimates Population Mean (μ)
- **Sample Proportion ($\hat{p}$)** → Estimates Population Proportion (p)
- **Sample Standard Deviation (s)** → Estimates Population Standard Deviation (σ)

2. Interval Estimation

An interval estimate provides a range of values (confidence interval) within which the population parameter is expected to lie with a certain degree of confidence (e.g., 95% confidence).

Example:

Continuing from the previous example, suppose we calculated a **95% confidence interval** for the average height of students to be **(158 cm, 162 cm)**. This means:

- We are **95% confident** that the **true average height (μ) of all students** lies between **158 cm and 162 cm**.

The formula for confidence interval of the mean:

$$CI = \bar{x} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

where:

- $\bar{x}$ = Sample mean
- $Z_{\alpha/2}$ = Z-score corresponding to the confidence level (e.g., 1.96 for 95%)
- σ = Population standard deviation (or sample standard deviation if unknown)
- n = Sample size

Why is Estimation Important?

1. **Saves Time & Resources:** Instead of surveying the whole population, we use a sample.
2. **Provides Decision-Making Insights:** Businesses, researchers, and policymakers rely on estimation.
3. **Allows for Margin of Error Consideration:** Confidence intervals show how precise our estimates are.

Large Sample Estimation of a Population Mean

Large sample estimation is used when the sample size (n) is **greater than or equal to 30** ($n \geq 30$). This allows us to apply the **Central Limit Theorem (CLT)**, which states that the sample mean ($\bar{x}$) follows a normal distribution, even if the population distribution is not normal.

Point Estimation for a Population Mean

A **point estimate** of the population mean (μ) is given by the **sample mean**:

$$\hat{\mu} = \bar{x}$$

where:

- $\bar{x}$ = Sample mean
- μ = Population mean (unknown and estimated)

Interval Estimation (Confidence Interval for Population Mean)

A confidence interval (CI) provides a range where the true population mean is likely to fall. When the sample size is large ($n \geq 30$), we use the **Z-distribution** (standard normal distribution) to calculate the confidence interval.

Formula for Confidence Interval (CI) of Population Mean

$$CI = \bar{x} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

where:

- $\bar{x}$ = Sample mean
- $Z_{\alpha/2}$ = Z-score for the confidence level (e.g., **1.96 for 95% confidence level**)
- σ = Population standard deviation (if known)
- n = Sample size

If the population standard deviation (σ) is **unknown**, we use the **sample standard deviation (s)** instead:

$$CI = \bar{x} \pm Z_{\alpha/2} \times \frac{s}{\sqrt{n}}$$

Example: Large Sample Estimation of Population Mean

Suppose a company wants to estimate the **average salary** of employees. They take a **random sample of 50 employees** ($n=50$), and find:

- **Sample mean salary** = $\bar{x} = 55,000$
- **Sample standard deviation** = $s = 5,000$
- **Confidence level** = 95% (which corresponds to $Z_{0.025} = 1.96Z$)

The **95% confidence interval** is calculated as:

$$\begin{aligned} CI &= 55,000 \pm \left(1.96 \times \frac{5000}{\sqrt{50}} \right) \\ &= 55,000 \pm (1.96 \times 707.11) \\ &= 55,000 \pm 1381.95 \\ &= (53,618.05, 56,381.95) \end{aligned}$$

Interpretation:

We are **95% confident** that the true average salary of all employees in the company lies between **\$53,618.05 and \$56,381.95\$**.

Small Sample Estimation of a Population Mean

Small sample estimation is used when the sample size $n < 30$ and the population standard deviation σ is unknown. Since the sample size is small, the **Central Limit Theorem (CLT)** does not **apply directly**, and we use the **t-distribution** instead of the normal (Z) distribution.

Point Estimation for a Population Mean

A **point estimate** of the population mean (μ) is given by the **sample mean**:

$$\hat{\mu} = \bar{x}$$

where:

- $\bar{x}$ = Sample mean
- μ = Population mean (unknown and estimated)

Interval Estimation (Confidence Interval for Population Mean)

Since the population standard deviation (σ) is unknown, we use the **sample standard deviation (s)** and the **t-distribution** (Student's t-distribution) to estimate the confidence interval.

Formula for Confidence Interval (CI) of Population Mean

$$CI = \bar{x} \pm t_{\alpha/2, (n-1)} \times \frac{s}{\sqrt{n}}$$

where:

- $\bar{x}$ = Sample mean
- $t_{\alpha/2, (n-1)}$ = t-score from the t-table with **(n – 1) degrees of freedom** and **confidence level $1 - \alpha$**
- s = Sample standard deviation
- n = Sample size

Example: Small Sample Estimation of Population Mean

A researcher wants to estimate the **average weight** of students in a university. She collects a **random sample of 10 students** ($n=10$) and finds:

- **Sample mean weight** = $\bar{x} = 65kg$
- **Sample standard deviation** = $s = 5 kg$
- **Confidence level** = 95%

From the **t-table**, for **n-1=9 degrees of freedom** and **95% confidence level**, the **t-value is 2.262**.

The **95% confidence interval** is:

$$\begin{aligned} CI &= 65 \pm \left(2.262 \times \frac{5}{\sqrt{10}} \right) \\ &= 65 \pm (2.262 \times 1.58) \\ &= 65 \pm 3.58 \\ &= (61.42, 68.58) \end{aligned}$$

Interpretation:

We are **95% confident** that the true average weight of all students in the university lies between **61.42 kg and 68.58 kg**.

Large Sample Estimation of a Population Proportion

In statistics, we often estimate a population **proportion (p)** when dealing with categorical data (e.g., the percentage of students who prefer online classes). **Large sample estimation of a population proportion** is used when the sample size is **large (n≥30)** and follows a normal distribution according to the **Central Limit Theorem (CLT)**.

Point Estimation for a Population Proportion

The best point estimate of a population proportion (p) is the sample proportion ($\hat{p}$):

$$\hat{p} = \frac{x}{n}$$

where:

- x = Number of individuals with the characteristic of interest
- n = Total sample size
- $\hat{p}$ = Sample proportion (point estimate of population proportion p)

Interval Estimation (Confidence Interval for Population Proportion)

To estimate the **true population proportion**, we calculate a **confidence interval (CI)** using the **normal (Z) distribution**.

Formula for Confidence Interval (CI) of Population Proportion

$$CI = \hat{p} \pm Z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where:

- $\hat{p}$ = Sample proportion
- $Z_{\alpha/2}$ = Z-score for the confidence level (e.g., **1.96 for 95% confidence level**)
- n = Sample size

Conditions for Large Sample Approximation (Normality Condition):

We can use the **normal distribution** for estimation if:

$$n\hat{p} \geq 5 \quad \text{and} \quad n(1 - \hat{p}) \geq 5$$

If these conditions are not met, a different method like the **exact binomial test** should be used.

Example: Large Sample Estimation of Population Proportion

A survey is conducted to estimate the proportion of people who **prefer electric cars**. A **random sample of 500 people** is selected, and **120 people** say they prefer electric cars.

Step 1: Calculate the Sample Proportion ($\hat{p}$)

$$\hat{p} = \frac{120}{500} = 0.24$$

Step 2: Choose Confidence Level (e.g., 95%)

For **95% confidence**, the **Z-score** is **1.96**.

Step 3: Compute the Confidence Interval

$$\begin{aligned} CI &= 0.24 \pm 1.96 \times \sqrt{\frac{0.24(1 - 0.24)}{500}} \\ &= 0.24 \pm 1.96 \times \sqrt{\frac{0.24 \times 0.76}{500}} \\ &= 0.24 \pm 1.96 \times \sqrt{0.0003648} \\ &= 0.24 \pm 1.96 \times 0.0191 \\ &= 0.24 \pm 0.0374 = (0.2026, 0.2774) \end{aligned}$$

Interpretation:

We are **95% confident** that the **true proportion of people who prefer electric cars** in the population lies between **20.26% and 27.74%**.

Sample Size Considerations

Choosing an appropriate **sample size (n)** is crucial for statistical analysis. A sample that is too small may lead to unreliable results, while an unnecessarily large sample increases costs and effort.

Factors Affecting Sample Size Selection

1. **Desired Confidence Level ($1 - \alpha$):**

- Higher confidence levels require **larger samples**.
- Example: A **99% confidence level** requires a larger sample than a **95% confidence level**.

2. **Margin of Error (E):**

- The **maximum allowable difference** between the sample estimate and the true population value.
- Smaller margins of error require **larger samples**.

3. **Population Variability (σ or $p(1 - p)$):**

- If the population has high variability, a **larger sample** is needed to capture diversity.

4. **Statistical Power:**

- The probability of correctly detecting a real effect.
- A higher power (e.g., 0.90 instead of 0.80) requires a **larger sample size**.

Formula for Sample Size Determination

For **estimating a population mean (μ)** when population **standard deviation (σ)** is known:

$$n = \left(\frac{Z_{\alpha/2} \times \sigma}{E} \right)^2$$

For **estimating a population proportion (p)**:

$$n = \frac{Z_{\alpha/2}^2 \times p(1 - p)}{E^2}$$

where:

- $Z_{\alpha/2}$ = Z-score for the confidence level
- σ = Population standard deviation (for mean estimation)
- p = Estimated population proportion (for proportion estimation)
- E = Margin of error

Example: Determining Sample Size for a Proportion

A researcher wants to estimate the proportion of students who use e-books instead of textbooks. She wants a **95% confidence level** and a **margin of error of 5% (0.05)**.

Using $Z_{0.025} = 1.96$ (for 95% confidence) and assuming $p=0.5$ (most conservative estimate),

$$\begin{aligned} n &= \frac{(1.96)^2 \times 0.5(1 - 0.5)}{(0.05)^2} \\ &= \frac{3.8416 \times 0.25}{0.0025} \\ &= \frac{0.9604}{0.0025} \\ &= 384.16 \end{aligned}$$

So, the researcher should **survey at least 385 students**.

Testing Hypotheses

Hypothesis testing is a statistical method used to make decisions about a population based on sample data.

Elements of Hypothesis Testing

1. Null Hypothesis (H_0):

- The assumption that there is **no effect** or **no difference**.
- Example: "The average height of students is 165 cm."

2. Alternative Hypothesis (H_a):

- The claim that there **is** an effect or difference.
- Example: “The average height of students **is not** 165 cm.”

3. Significance Level (α):

- The probability of rejecting H_0 when it is actually true (Type I error).
- Common values: **0.05 (5%)** or **0.01 (1%)**.

4. Test Statistic:

- A formula used to compare sample data with the hypothesis.
- For **large samples** (when $n \geq 30$), the **Z-test** is used.
- For **small samples** (when $n < 30$), the **t-test** is used.

5. P-value:

- The probability of obtaining the observed result under H_0 .
- If $p < \alpha$, reject H_0 (strong evidence against it).

6. Decision Rule:

- If the **test statistic** falls in the rejection region, we reject H_0 .

Example: Hypothesis Testing for a Mean (Large Sample)

A company claims that its batteries last **at least 50 hours**. A researcher tests **100 batteries** and finds an **average life of 48.5 hours** with a **standard deviation of 4 hours**.

- H_0 : The average battery life is **at least 50 hours** ($\mu \geq 50$).
- H_a : The average battery life is **less than 50 hours** ($\mu < 50$).

Step 1: Compute Test Statistic

For large samples ($n \geq 30$), use the **Z-test**:

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$
$$= \frac{48.5 - 50}{4 / \sqrt{100}}$$

$$= \frac{-1.5}{0.4}$$

$$= -3.75$$

Step 2: Determine Critical Value

For a **one-tailed test** at $\alpha=0.05$, the **Z-critical value** is **-1.645**.

Step 3: Compare Z-score with Critical Value

Since $-3.75 < -1.645$, we **reject H_0** .

Conclusion:

There is **strong evidence** that the average battery life is **less than 50 hours**.

The Elements of Hypothesis Testing

Hypothesis testing is a fundamental statistical technique used to make decisions about a population based on sample data. It helps determine whether there is enough evidence to support a particular claim.

1. Null Hypothesis (H_0)

- The **null hypothesis** represents the **status quo** or an assumption that there is **no effect** or **no difference** in the population.
- It is denoted as H_0 .
- Example: A pharmaceutical company claims that its new drug has the **same effect** as the old one. $H_0: \mu = \mu_0$ (where μ_0 is the population mean based on the existing drug)

2. Alternative Hypothesis (H_a or H_1)

- The **alternative hypothesis** contradicts the null hypothesis. It suggests that there is a **significant effect** or **difference** in the population.
- It is denoted as H_a or H_1 .
- Example: The new drug is **more effective** than the old one. $H_a: \mu > \mu_0$

Types of Alternative Hypotheses:

- **One-tailed test (directional):** Tests if the parameter is either greater or smaller than a certain value.

- Example: $H_a: \mu > \mu_0$ (right-tailed) or $H_a: \mu < \mu_0$ (left-tailed)
- **Two-tailed test (non-directional):** Tests if the parameter is **different** from a certain value.
 - Example: $H_a: \mu \neq \mu_0$

3. Significance Level (α)

- The **significance level** (α) represents the **probability of rejecting H_0 when it is actually true** (Type I error).
- Common values: **0.05 (5%), 0.01 (1%), or 0.10 (10%)**.
- A lower α means **stricter evidence is required** to reject H_0 .
- ◆ **Example:** If $\alpha = 0.05$, there is a **5% chance** of incorrectly rejecting H_0 when it is true.

4. Test Statistic

- The **test statistic** measures how far the sample data deviates from the null hypothesis.
- The type of test statistic depends on the sample size and population characteristics:
 - **Z-test** (for large samples, $n \geq 30$, or when population variance is known)
 - **t-test** (for small samples, $n < 30$, and unknown population variance)
- ◆ **Formula for Z-test (Large Sample, Mean Known):**

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

- $\bar{x}$ = Sample mean
- μ_0 = Population mean (under H_0)
- σ = Population standard deviation
- n = Sample size

- ◆ **Formula for t-test (Small Sample, Variance Unknown):**

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

- s = Sample standard deviation

5. P-Value (Probability Value)

- The **p-value** is the probability of observing a test statistic **as extreme or more extreme** than the one obtained, assuming H_0 is true.
- **Decision rule based on p-value:**
 - If $p \leq \alpha \rightarrow$ Reject H_0 (**Strong evidence against H_0**)
 - If $p > \alpha \rightarrow$ Fail to reject H_0 (**Not enough evidence**)

♦ **Example:** If $p=0.03$ and $\alpha=0.05$, then **reject H_0** because $0.03 < 0.05$.

6. Decision Rule

- Based on the **test statistic** and **critical value** (from statistical tables), we make a decision:
 - If **test statistic falls in the rejection region**, we **reject H_0** .
 - If **test statistic falls in the acceptance region**, we **fail to reject H_0** .

♦ **Example (One-Tailed Test):**

A university claims that students spend **at least 10 hours per week** studying. A random sample of 40 students shows a **mean study time of 9.5 hours** with a **standard deviation of 2 hours**. We test:

$H_0: \mu \geq 10$ vs. $H_a: \mu < 10$ (one-tailed, left)

- Significance level: $\alpha=0.05$
- Compute test statistic:

$$Z = \frac{9.5 - 10}{2/\sqrt{40}} = \frac{-0.5}{0.316} = -1.58$$

- Critical value at $\alpha=0.05$ (one-tailed) is **-1.645**.
- Since **-1.58 > -1.645**, we **fail to reject H_0** .
- Conclusion: **Not enough evidence** to say students study less than 10 hours.

Large Sample Tests for a Population Mean

A **large sample test for a population mean** is used when the sample size is **greater than or equal to 30** ($n \geq 30$). The test helps determine whether the sample mean significantly differs from a claimed population mean.

1. When to Use a Large Sample Test?

- When the sample size is **large** ($n \geq 30$).
- When the population **standard deviation** (σ) is **known** (Z-test).
- If σ is unknown, we approximate it using the sample standard deviation (s).
- The test follows a **normal distribution (Z-distribution)** based on the Central Limit Theorem.

2. Hypotheses in Large Sample Mean Tests

We test a claim about the population mean (μ) using:

- **Null Hypothesis (H_0)**: The population mean equals a specified value. $H_0: \mu = \mu_0$
- **Alternative Hypothesis (H_a)**: The population mean is different, greater, or smaller than the specified value.
- $H_a: \mu \neq \mu_0$ (two-tailed test)
- $H_a: \mu > \mu_0$ (right-tailed test)
- $H_a: \mu < \mu_0$ (left-tailed test)

3. Test Statistic for Large Sample Mean (Z-test)

For a large sample ($n \geq 30$), we use the **Z-test statistic**:

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Where:

- $\bar{x}$ = Sample mean
- μ_0 = Population mean (hypothesized)
- σ = Population standard deviation
- n = Sample size

If the population standard deviation σ is unknown, we use the sample standard deviation s instead:

$$Z = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

4. Decision Rule

1. **Set the significance level (α)**, commonly **0.05 (5%)** or **0.01 (1%)**.
2. **Find the critical value (Z_α)** from the standard normal table:
 - For a **two-tailed test** at $\alpha = 0.05$, critical values are **± 1.96** .
 - For a **one-tailed test** at $\alpha = 0.05$, the critical value is **1.645 (right) or -1.645 (left)**.
3. **Compare the test statistic Z with the critical value:**
 - If $|Z| > Z_\alpha$, **reject H_0** (strong evidence against H_0).
 - If $|Z| \leq Z_\alpha$, **fail to reject H_0** (not enough evidence).

5. Example: Testing the Mean Study Hours of Students

A university claims that students spend **at least 10 hours per week** studying. A researcher surveys **50 students** and finds an **average study time of 9.5 hours** with a **standard deviation of 2.5 hours**. Test at $\alpha = 0.05$.

Step 1: Define Hypotheses

- $H_0: \mu \geq 10$ (Students study at least 10 hours)
- $H_a: \mu < 10$ (Students study less than 10 hours)

Step 2: Compute the Z-test Statistic

Given:

- $\bar{x} = 9.5$
- $\mu_0 = 10$
- $s = 2.5$
- $n = 50$

Using the formula:

$$Z = \frac{9.5 - 10}{2.5/\sqrt{50}} = \frac{-0.5}{2.5/7.07} = \frac{-0.5}{0.354} = -1.41$$

Step 3: Find Critical Value

- For a **one-tailed test** at $\alpha = 0.05$, the critical value is **-1.645**.

Step 4: Compare and Make a Decision

- Since $Z = -1.41$ is **greater than** -1.645 , we **fail to reject** H_0 .
- Conclusion: There is **not enough evidence** to say that students study less than 10 hours per week.

6. Conclusion

Large sample tests for a population mean are **used when $n \geq 30$** and help determine whether the sample mean significantly differs from a hypothesized value. The **Z-test** is the preferred method, and the decision is made by comparing the test statistic with a critical value or using the p-value approach.

The Observed Significance of a Test (p-value Approach)

The **observed significance level**, also known as the **p-value**, is the probability of obtaining a test statistic as extreme as the one observed, assuming that the **null hypothesis (H_0) is true**.

Instead of comparing a test statistic to a critical value, the **p-value approach** provides a direct measure of evidence against H_0 .

1. Definition of p-value

- The **p-value** represents the **smallest level of significance** at which the null hypothesis can be rejected.
- It measures how likely we are to get a sample result **as extreme or more extreme** than the observed one.
- A **small p-value** (e.g., less than 0.05) indicates **strong evidence against H_0** , suggesting we should reject it.

2. Decision Rule using p-value

1. **Set the significance level α** (common values: **0.05 or 0.01**).
2. **Calculate the test statistic** (e.g., Z-test or t-test).
3. **Find the p-value** corresponding to the test statistic.
4. **Compare the p-value with α :**
 - If **$p\text{-value} \leq \alpha \rightarrow \text{Reject } H_0$** (Strong evidence against H_0)
 - If **$p\text{-value} > \alpha \rightarrow \text{Fail to reject } H_0$** (Not enough evidence against H_0)

3. Interpretation of p-value

- **p-value < 0.01** → **Very strong evidence** against H_0 (reject H_0).
- **p-value between 0.01 and 0.05** → **Strong evidence** against H_0 (reject H_0).
- **p-value between 0.05 and 0.10** → **Weak evidence** against H_0 (borderline case).
- **p-value > 0.10** → **No strong evidence** against H_0 (fail to reject H_0).

4. Example: Testing the Mean Study Hours of Students

A researcher claims that students study **10 hours per week on average**. A sample of **50 students** gives a **mean of 9.5 hours** with a **standard deviation of 2.5 hours**. Test at $\alpha = 0.05$.

Step 1: Hypotheses

- $H_0: \mu = 10$ (Students study 10 hours per week)
- $H_a: \mu \neq 10$ (Students study a different amount)

Step 2: Compute Test Statistic (Z-test)

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
$$Z = \frac{9.5 - 10}{2.5/\sqrt{50}}$$
$$Z = \frac{-0.5}{0.354} = -1.41$$

Step 3: Find p-value

- Since this is a **two-tailed test**, we find the probability $P(Z \leq -1.41)$ from the standard normal table.
- Looking at the **Z-table**, $P(Z \leq -1.41) = 0.0793$.
- Since this is a **two-tailed test**, we multiply by 2:

$$p - value = 2 \times 0.0793 = 0.1586$$

Step 4: Compare p-value with α

- Given $\alpha = 0.05$ and **p-value = 0.1586**, we see that **p-value > 0.05**.
- **Conclusion:** Since the p-value is greater than 0.05, we **fail to reject H_0** . There is **not enough evidence** to say that the study hours are different from 10 hours.

5. Conclusion

The **p-value** provides a **quantitative measure** of evidence against the null hypothesis. If the p-value is **small**, we reject H_0 . If it is **large**, we fail to reject H_0 , meaning we do not have enough evidence to prove the alternative hypothesis.

Small Sample Tests for a Population Mean

When the sample size is **small** ($n < 30$), and the population **standard deviation (σ) is unknown**, we use the **t-test** instead of the **Z-test**. This test follows the **Student's t-distribution**, which accounts for the additional variability in small samples.

1. When to Use a Small Sample Test?

- The **sample size is small** ($n < 30$).
- The **population standard deviation (σ) is unknown**.
- The population is assumed to be **normally distributed** or approximately normal.
- We use the **t-distribution** instead of the normal (Z) distribution.

2. Hypotheses in a Small Sample Mean Test

We test a claim about the population mean (μ) using:

- **Null Hypothesis (H_0)**: The population mean equals a specified value. $H_0: \mu = \mu_0$
- **Alternative Hypothesis (H_a)** (depending on the claim):
 - Two-tailed: $H_a: \mu \neq \mu_0$ (mean is different).
 - Right-tailed: $H_a: \mu > \mu_0$ (mean is greater).
 - Left-tailed: $H_a: \mu < \mu_0$ (mean is smaller).

3. Test Statistic for Small Sample Mean (t-test)

For a **small sample** ($n < 30$), we use the **t-test statistic**:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Where:

- $\bar{x}$ = Sample mean
- μ_0 = Population mean (hypothesized)
- s = Sample standard deviation
- n = Sample size

This statistic follows a **t-distribution** with $n - 1$ **degrees of freedom (df)**.

4. Decision Rule

1. **Set the significance level (α)**, commonly **0.05 (5%)** or **0.01 (1%)**.
2. **Find the critical t-value ($t_{\alpha, df}$)** from the **t-table** using $df = n - 1$.
3. **Compare the test statistic ttt with the critical t-value:**
 - If $|t| > t_{\alpha, df} \rightarrow$ **Reject H_0** (strong evidence against H_0).
 - If $|t| \leq t_{\alpha, df} \rightarrow$ **Fail to reject H_0** (not enough evidence).

5. Example: Testing the Effect of a New Teaching Method

A professor claims that students taught using a new method score an average of **75** on a test. A sample of **10 students** gave the following scores:

72, 78, 80, 76, 74, 71, 79, 77, 73, 75

At $\alpha = 0.05$, test whether the mean score is **different** from 75.

Step 1: Define Hypotheses

- $H_0: \mu = 75$ (The average score is 75).
- $H_a: \mu \neq 75$ (The average score is different).

Step 2: Compute Sample Statistics

- **Sample mean ($\bar{x}$):**
 - $\bar{x} = \frac{72+78+80+76+74+71+79+77+73+75}{10} = \frac{755}{10} = 75.5$
- **Sample standard deviation (s):**

Using the formula: $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$

After calculations, $s = 3.14$.

Step 3: Compute t-test Statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
$$t = \frac{75.5 - 75}{3.14/\sqrt{10}}$$
$$t = \frac{0.5}{0.993} = 0.50$$

Step 4: Find Critical t-value

- **Degrees of freedom:** $df = n - 1 = 10 - 1 = 9$.
- From the **t-table**, for a **two-tailed test** at $\alpha = 0.05$ and $df = 9$, the **critical t-value** is **2.262**.

Step 5: Compare t-value with Critical Value

- Since $|0.50| < 2.262$, we **fail to reject H_0** .

Step 6: Conclusion

There is **not enough evidence** to conclude that the new teaching method produces a different average test score from 75.

Large Sample Tests for a Population Proportion

When dealing with **large sample sizes** in hypothesis testing for a **population proportion (ppp)**, we use the **Z-test**. This test follows the **normal distribution** and helps determine whether the sample proportion ($\hat{p}$) is significantly different from a hypothesized population proportion (p_0).

1. When to Use a Large Sample Test for Proportion?

We use this test when:

- The **sample size is large** ($n > 30$).
- The number of **successes** and **failures** satisfy: $np_0 \geq 5$ and $n(1 - p_0) \geq 5$
- The sample is **randomly selected**.

- The sampling distribution of $\hat{p}$ (sample proportion) follows a normal distribution.

2. Hypotheses for Proportion Tests

We test a claim about the **population proportion (p)**:

- **Null Hypothesis (H_0)**: The population proportion equals a specified value. $H_0: p = p_0$
- **Alternative Hypothesis (H_a)**:
 - **Two-tailed**: $H_a: p \neq p_0$ (proportion is different).
 - **Right-tailed**: $H_a: p > p_0$ (proportion is greater).
 - **Left-tailed**: $H_a: p < p_0$ (proportion is smaller).

3. Test Statistic for Proportion (Z-test Formula)

For **large samples**, the test statistic follows a **standard normal distribution (Z-distribution)**:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

Where:

- $\hat{p}$ = Sample proportion = $\frac{x}{n}$ (where x is the number of successes, and n is the sample size)
- p_0 = Hypothesized population proportion
- n = Sample size

4. Decision Rule

1. **Set the significance level (α)**, usually **0.05 (5%)** or **0.01 (1%)**.
2. **Find the critical Z-value (Z_α)** from the **Z-table**:
 - **For two-tailed test**: $Z_{\alpha/2}$.
 - **For one-tailed test**: Z_α .
3. **Compare the test statistic Z with the critical Z-value**:
 - If $|Z| > Z_{\alpha/2} \rightarrow$ **Reject H_0** .
 - If $|Z| \leq Z_{\alpha/2} \rightarrow$ **Fail to reject H_0** .

5. Example: Testing the Proportion of Defective Items

A factory claims that **5%** of its products are defective. A quality control team inspects **400 items** and finds **30 defective ones**. Test at $\alpha = 0.05$ if the defect rate is **higher** than 5%.

Step 1: Define Hypotheses

- $H_0: p = 0.05$ (Defect rate is 5%).
- $H_a: p > 0.05$ (Defect rate is higher than 5%).

Step 2: Compute Sample Proportion

$$\hat{p} = \frac{x}{n} = \frac{30}{400} = 0.075$$

Step 3: Compute Z-test Statistic

$$\begin{aligned} Z &= \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \\ Z &= \frac{0.075 - 0.05}{\sqrt{\frac{0.05(1-0.05)}{400}}} \\ Z &= \frac{0.025}{\sqrt{\frac{0.05 \times 0.95}{400}}} \\ Z &= \frac{0.025}{\sqrt{\frac{0.0475}{400}}} \\ Z &= \frac{0.025}{\sqrt{0.00011875}} = \frac{0.025}{0.0109} = 2.29 \end{aligned}$$

Step 4: Find Critical Z-value

- Since it is a **right-tailed test** at $\alpha = 0.05$,
 $Z_{0.05} = 1.645$ (from Z-table).

Step 5: Compare Z-value with Critical Value

- Since $2.29 > 1.645$, we reject H_0 .

Step 6: Conclusion

There is **significant evidence** that the defect rate is **higher than 5%**.